Generating Functions 4

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Problem 1

This can be thought of as a sequence of four sets of objects where the 3rd set has an odd number of objects and the 4th set has an even number of objects. So the specification is

$$\mathrm{SET}(\mathcal{Z}) \star \mathrm{SET}(\mathcal{Z}) \star \mathrm{SET}_{\mathrm{even}}(\mathcal{Z}) \star \mathrm{SET}_{\mathrm{odd}}(\mathcal{Z}),$$

and the generating function is

$$e^{2z}(\sinh z)(\cosh z) = \frac{1}{2}e^{2z}\sinh(2z) = \frac{1}{4}(e^{4z}-1).$$

The count is then

$$n![z^n]\frac{1}{4}(e^{4z}-1) = 4^{n-1}.$$

Problem 3

We have that

$$A'(z) = \frac{d}{dz}\sum_{n\geq 0} a_n \frac{z^n}{n!} = \frac{d}{dz}\sum_{n\geq 0} a_{n+1}\frac{z^{n+1}}{(n+1)!} = \sum_{n\geq 0} a_{n+1}\frac{z^n}{n!},$$

which is the desired EGF, so the answer is $A'(z)$.

Problem 4

Expanding the RHS, we get

$$\int_0^\infty e^{-t}\widehat{A}(tz)dt = \int_0^\infty e^{-t}\sum_{n\geq 0} a_n\frac{(tz)^n}{n!}dt = \sum_{n\geq 0}\int_0^\infty e^{-t}a_n\frac{t^n z^n}{n!}dt$$

$$= \sum_{n\geq 0} a_n\frac{z^n}{n!}\int_0^\infty e^{-t}t^n dt = \sum_{n\geq 0} a_n\frac{z^n}{n!}n! = \sum_{n\geq 0} a_n z^n = A(z),$$

as desired.

Problem 5

We can think of a ranking as a sequence of ranks (favorite to least favorite), where each rank has a set of objects. This is the same structure as a surjection, so the EGF is

$$\frac{1}{2-e^z}.$$

Problem 6

The specification is

$$\mathrm{SET}(\mathrm{SET}_{\geq 2}(z)),$$

so the generating function is

$$e^{e^z-1-z}.$$

Problem 8

The specification is

$$\mathrm{SET}(\mathrm{SET}_{\geq 2}(z)),$$

so the generating function is

$$\frac{1}{1-(e^z-1-z)} = \frac{1}{2-z-e^z}.$$

Problem 12

These arrangements are equivalent to a cycle of atoms and a single atom, so the specification is

$$\mathrm{CYC}(\mathcal{Z}) \star \mathcal{Z},$$

and the generating function is

$$z \log \frac{1}{1-z}.$$

Problem 13

Since we're taking a set partition, then partitioning each of its blocks again, the result is an unordered set of blocks of a set partition. So the generating function is

$$C(z) = e^{e^{e^z-1}-1}.$$

Problem 14

The specification is

$$\mathrm{SET}(\mathrm{CYC}_{\text{divisible by } k}(\mathcal{Z})).$$

So the EGF is

$$\exp\left(\sum_{j\geq 1}\frac{z^{kj}}{kj}\right) = \exp\left(\frac{1}{k}\log\left(\frac{1}{1-z^k}\right)\right) = \frac{1}{(1-z^k)^{1/k}}.$$

So

$$p_n = n![z^n](1-z^k)^{-1/k}.$$

To extract the z^n coefficient, we use the generalized binomial theorem:

$$(1-z^k)^{-1/k} = \sum_{j\geq 0}\binom{-1/k}{j}(-1)^j z^{kj}.$$

Simplifying the coefficient, we get

$$(-1)^j\binom{-1/k}{j} = \frac{(-1)^j}{j!}\left(-\frac{1}{k}\right)\left(-\frac{1}{k}-1\right)\cdots\left(-\frac{1}{k}-j+1\right) = \frac{1}{j!}\prod_{i=0}^{j-1}\left(\frac{1}{k}+i\right).$$

Therefore, the EGF is

$$\sum_{j\geq 0}\frac{z^{kj}}{j!}\prod_{i=0}^{j-1}\left(\frac{1}{k}+i\right),$$

so

$$p_n = \frac{n!}{(n/k)!}\prod_{i=0}^{n/k-1}\left(\frac{1}{k}+i\right),$$

when $k \mid n$, and 0 otherwise.

Problem 16

A permutation has a square root if all its cycles of even length come in pairs of the same length. So we can make a valid permutation from a set of odd cycles along with sets (of even length) of cycles of even length. So the specification is

$$\mathrm{SET}(\mathrm{CYC}_{\mathrm{odd}}(\mathcal{Z})) \star \prod_{k \text{ even}} \mathrm{SET}_{\mathrm{even}}(\mathrm{CYC}_k(\mathcal{Z})).$$

The odd part has generating function

$$\exp\left(\sum_{j\geq 0}\frac{z^{2j+1}}{2j+1}\right).$$

For each even k, since $\mathrm{SET}_{\mathrm{even}}(z) = \cosh(z)$, the generating function for the even part is

$$\prod_{j\geq 1} \cosh\left(\frac{z^{2j}}{2j}\right),$$

so the final generating function is

$$\exp\left(\sum_{j\geq 0} \frac{z^{2j+1}}{2j+1}\right) \prod_{j\geq 1} \cosh\left(\frac{z^{2j}}{2j}\right).$$